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ABSTRACT

The incompressible Falkner–Skan–Cooke (FSC) boundary layer is extended to the compressible regime on a flat plate, aiming at establishing a generalized model to facilitate the crossflow-related investigations. The compressible FSC flow is also compatible with the two-dimensional compressible Blasius boundary layer characterized by a self-similarity solution. A family of boundary layer with local self-similarity is obtained from an ordinary differential equation system. Pressure-gradient parameter and local swept angle are introduced to parameterize the crossflow, which can vary in both direction and magnitude. The results reveal that the crossflow is oriented toward the curvature center of the inviscid streamline, regardless of the direction of pressure gradient. The existence of crossflow results in moderate directional deviation of skin friction, while tiny difference is observed for the magnitude. In comparison, much less effect is observed on the thermal properties of boundary layer. Crossflow also leads to the distortion of vorticity inside boundary layer, accompanied by the production of helicity. In addition, it is revealed that cross-flow strength varies monotonously with the pressure-gradient parameter, while it varies non-monotonously with the local swept angle. The effects of Mach number and wall temperature on crossflow are investigated as well.

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NOMENCLATURE

a	speed of sound	T_s	temperature factor in Sutherland's law
C	constant coefficient of Falkner–Skan velocity	(U, V, W)	velocity vector after Illingworth–Stewartson transformation
C_f	skin-friction coefficient	(u, v, w)	velocity vector in Cartesian coordinate system
c_p	specific heat capacity	$(\bar{u}, \bar{v}, \bar{w})$	velocity vector in stream-fitted coordinate system
f	wall-normal part of stream function	(x, y, z)	Cartesian coordinates
g	scaled velocity in z direction	$(\bar{x}, \bar{y}, \bar{z})$	stream-fitted coordinates
h	helicity	γ	ratio of specific heats
K	combination parameter in energy equation	δ	boundary-layer thickness
L	combination parameter in velocity expressions	θ	skin friction
l	boundary-layer length scale	κ	thermal conductivity
m	pressure-gradient parameter	Λ	local swept angle at the reference point
Ma	Mach number	Λ_1	local swept angle at a given streamwise point
N	Chapman–Rubesin factor	μ	dynamic viscosity
p	pressure	ν	kinematic viscosity
Pr	Prandtl number	(ξ, η)	coordinates after Falkner–Skan transformation
q	heat transfer	$(\bar{\xi}, \bar{\eta})$	coordinates after Illingworth–Stewartson transformation
Re	Reynolds number	ρ	density
S	combination parameter in energy equation	τ	scaled temperature
St	Stanton number	φ	deviation angle
T	temperature	χ	combination parameter in energy equation
		ψ	stream function
		ω	vorticity

Subscripts

<i>ad</i>	adiabatic wall
<i>cf</i>	crossflow
<i>cr</i>	crosswise direction
<i>e</i>	boundary-layer edge
<i>p</i>	quantities when crossflow peaks
<i>ref</i>	reference point
<i>str</i>	streamwise direction
<i>w</i>	wall
∞	quantities in external stream
0	stagnation condition

I. INTRODUCTION

Crossflow is the secondary flow created by pressure gradient perpendicular to streamwise direction in three-dimensional (3D) boundary layers. Crossflow appears widely on practical flight vehicles with 3D configurations, such as swept wing,^{1,2} elliptic cone,³ and yawed circular cone.^{4,5} Rotating body of revolution is another type of configuration where crossflow is commonly encountered.^{6,7} Generally, the cross-flow velocity owns a C-shape profile, which is regarded as the source of cross-flow instability and transition. Cross-flow vortices, stationary or traveling, are unique in this type of transition process. Compressible boundary-layer transition gives rise to sharp increases of drag and heat transfer, especially in hypersonic velocity regime; thus, the prediction and control are of primary concerns in engineering practice. As owning an exclusive mechanism, crossflow-induced transition receives more and more interest and attention.^{8–14}

Different geometries have been considered to explore the mechanisms of crossflow-related phenomena. Bippes¹⁵ reviewed experimental studies related to cross-flow instability, highlighting the increased role of crossflow-dominated transition on various geometries. Saric *et al.*⁸ made a comprehensive review about cross-flow instability in incompressible and weakly compressible boundary layers. Combined with the consideration of research related to moderately and highly compressible conditions, 3D boundary layers can be categorized into three groups. The first one is the boundary layer on a curved surface. This group might also be the most widely encountered and investigated. For example, Balakumar and King¹ considered a swept cylinder and obtained the supersonic boundary layer using CFL3D code. Ustinov and Kachanov¹⁶ generated cross-flow instability modes by surface roughness on a swept airfoil. Moyes *et al.*⁴ analyzed the secondary instability of crossflow in a hypersonic boundary layer on a yawed circular cone, with a half-angle 7° and an angle of attack (AoA) 6° . A so-called DPLR was adopted to resolve the laminar boundary layer. Wan *et al.*¹⁷ also considered the yawed cone and identified the cross-flow waves in experiments. In this group, cross-flow effect is not exclusive in the boundary layers, in which wall curvature, body-shape parameters, and inclination are inevitably affecting flow physics, and the crossflow can hardly be regulated by such factors. The second group is the boundary layer on rotating bodies, like disk,¹ cone,⁷ and sphere.¹⁸ In this particular group, neither of the streamwise nor crosswise component of velocity vanishes at the wall, which results in unique profiles differing from the majority of 3D boundary layers. Besides, a highly compressible boundary

layer often implies a crossflow with high speed, which might be hard to generate by rotation in practical applications. The third group is the boundary layer on a fixed flat plate with a swept leading edge. For a subsonic incoming flow, a displacement body is usually mounted over the plate to create the desired pressure gradients.^{19,20} While for a supersonic incoming flow, in order to create crossflow, the leading edge is usually bluntly curved in experiment²¹ and numerical simulations,^{22,23} or a non-uniformly distributed velocity in free stream is assigned.^{9,24,25} In a nutshell, a curved geometry, or a flat plate with prescribed potential flow, is commonly considered to study the cross-flow effects in a compressible boundary layer.

Although the boundary layer on the complex geometries can be tackled directly with the aid of better- and better-developed numerical tools, the lack of a model problem is still believed to be impeding the exploration of the basic aspects of crossflow. The reason lies in three aspects. First, on the practical configurations, the cross-flow effect is coupled with other effects of the factors listed previously. Thus, the result of cross-flow effects is inevitably contaminated. Second, the uniqueness of certain geometry results in the cross-flow effects with particularity instead of generality. The geometry-dependent feature disguises the fundamentals related to crossflow and undermines the justification to make extension from case to case. Third, it costs more time and computation resources to resolve the boundary layer on complex geometries. Besides, the numerical errors may also lead to issues about validation and replication. Therefore, a simple configuration is preferred in building a compressible flow model for the crossflow-related investigations. A notably good example had been made by Mack,²⁶ who conducted a comprehensive and fundamental research about linear stability of two-dimensional (2D) compressible boundary layers based on a flat-plate flow model which admits an exact solution. The Falkner–Skan–Cooke (FSC) flow model has also been utilized in developing a cross-flow transition model based on local variables.²⁷

The attempts to build a 3D boundary-layer model can be dated back to 1950s. Cooke²⁸ obtained an ordinary differential equation (ODE) system for the incompressible boundary layers on yawed cylinders. The Falkner–Skan velocity distribution was adopted and thereby the flow has been known as Falkner–Skan–Cooke (FSC) flow. Cohen and Reshotko²⁹ and Reshotko and Beckwith^{30,31} considered compressible boundary layers with special attention paid to the assumptions under which there exist self-similar solutions. The assumptions were revealed to include linear law of viscosity, unity Prandtl number, and unity specific heat ratio. Such assumptions, which may not cause unacceptable differences in estimating the skin friction and heat transfer, are not qualified to investigate the crossflow-related mechanisms, like cross-flow instability. After dropping the aforementioned assumptions, a numerical method is usually necessary to obtain the solution of a non-similar boundary layer. Gaponov and Smorodskii³² proposed a quasi-ODE system to study the supersonic cross-flow instability. The system can be solved locally, while unfortunately, an error occurs on the right-hand side of the energy equation. Tempelmann *et al.*²⁴ proposed a similar system on a flat plate with the solution as starting profiles for the partial differential equation (PDE) system. Xu *et al.*⁹ also considered a Falkner–Skan potential flow, while the fully boundary-layer equation is solved numerically. Although these attempts result in encouraging results, a well-accepted model has not been achieved for compressible boundary layer with crossflow.

The present work builds a model flow of compressible boundary layer which may serve to facilitate fundamental research related to crossflow. With prescribed pressure gradient over a flat plate, a swept flow produces a boundary layer with crossflow. On the one hand, fewest parameters, i.e., the pressure-gradient parameter and local swept angle, are used to establish the simplest model by neglecting unnecessary effects like wall curvature. The two parameters are also responsible for creating crossflows with varying directions and magnitudes. On the other hand, an ODE system is obtained to further simplify the boundary-layer problem with assumption of local self-similarity. The presented model flow is the generalization of the incompressible FSC boundary layer and 2D compressible Blasius boundary layer. Therefore, boundary-layer characteristics can be bridged between well-known flows and 3D compressible flows. Basic properties are presented in the current work, including velocity profiles, skin friction, vorticity, and helicity distributions. The variations of boundary-layer properties are also investigated under the effects of pressure gradient, swept angle, Mach number, and wall temperature. Conclusions are made at the end.

II. FORMULATION OF 3D BOUNDARY LAYER

A. Model problem

On an infinite flat plate, the Cartesian coordinate system is established with x and z directions parallel to the plate and y direction normal to the wall (Fig. 1). An inviscid stream flows in an inclined direction with velocity component U_e in the x direction and W_e in the z direction. Generally, U_e can be zero, while W_e is set to be non-zero. A pressure gradient is assigned in the x -direction in the region $x > 0$. Under such pressure gradient, the flow accelerates or decelerates in the x direction with a Falkner–Skan velocity component. The angle between velocity components in x and z directions is named local swept angle, denoted as Λ . The inviscid streamline skews from initial direction and gets curved, accompanied by a varying Λ downstream.

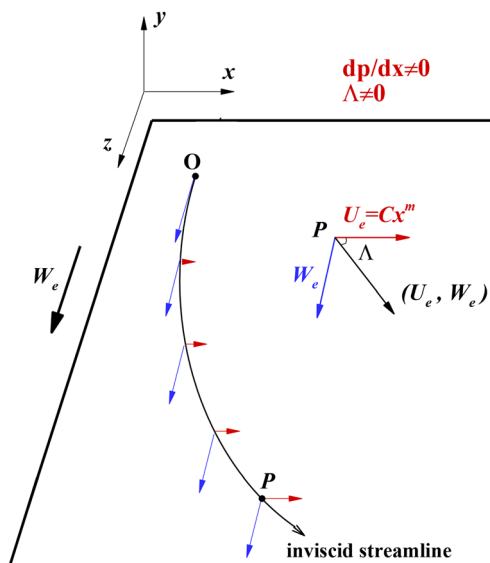


FIG. 1. Sketch of 3D boundary layer on a flat plate.

B. Governing equations

Consider the boundary layer attached to the plate as shown in Fig. 1. Although the velocity contains three components, the boundary layer is assumed to be homogeneous in the z direction. Thus, the variations in z direction are not considered. Therefore, the compressible 3D boundary-layer equations are introduced as follows.³³

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0, \quad (1)$$

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \rho_e u_e \frac{du_e}{dx} + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right), \quad (2)$$

$$\rho \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} \right) = \frac{\partial}{\partial y} \left(\mu \frac{\partial w}{\partial y} \right), \quad (3)$$

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) + u \rho_e u_e \frac{du_e}{dx} = \frac{\partial}{\partial y} \left(\kappa \frac{\partial T}{\partial y} \right) + \mu \left(\frac{\partial u}{\partial y} \right)^2 + \mu \left(\frac{\partial w}{\partial y} \right)^2, \quad (4)$$

in which perfect gas is assumed and μ is dynamic viscosity, κ thermal conductivity, and c_p specific heat capacity. The subscript e denotes the edge of boundary layer. The variables ρ , u , v , w , and T represent density, x -direction velocity, y -direction velocity, z -direction velocity, and temperature, respectively. The dynamic viscosity μ is calculated using the Sutherland's law.

No-slip and constant/adiabatic temperature conditions are prescribed at the wall. The velocity and temperature at the boundary-layer edge can be obtained through isentropic relations. Thus, the boundary conditions are given as

$$y = 0 : u = v = w = 0, \quad T = T_w \quad \text{or} \quad \frac{\partial T}{\partial y} = 0, \quad (5)$$

$$y \rightarrow \infty : u \rightarrow u_e, v \rightarrow 0, w \rightarrow w_e, \quad T \rightarrow T_e, \quad (6)$$

in which the subscript w denotes the wall.

Based on the continuity equation Eq. (1), the stream function ψ is defined by^{30,34}

$$\frac{\partial \psi}{\partial y} = \frac{\rho u}{\rho_{e0}}, \quad \frac{\partial \psi}{\partial x} = -\frac{\rho v}{\rho_{e0}}, \quad (7)$$

where subscript "0" denotes a stagnation quantity.

In order to remove the density ρ from governing equations, Illingworth–Stewartson transformation³⁴ is introduced as

$$d\bar{\xi} = \frac{\mu_e a_e \rho_e}{\mu_{e0} a_{e0} \rho_{e0}} dx, \quad d\bar{\eta} = \frac{a_e \rho}{a_{e0} \rho_{e0}} dy, \quad (8)$$

in which a is sound speed. In the transformed coordinate system, the velocity components are defined as²⁹

$$\frac{\partial \psi}{\partial \bar{\eta}} = U, \quad \frac{\partial \psi}{\partial \bar{\xi}} = -V. \quad (9)$$

Combining Eqs. (7)–(9), one can obtain

$$u = \frac{a_e}{a_{e0}} U, \quad v = \frac{\mu_e a_e \rho_e}{\mu_{e0} a_{e0} \rho_{e0}} V - \frac{\rho_{e0}}{\rho} U \frac{\partial \bar{\eta}}{\partial \bar{\xi}}. \quad (10)$$

The velocities in the two coordinate systems are related through Eq. (10). As a matter of fact, some other transformations can usually

be adopted, like Lees–Dorodnitsyn transformation.³⁵ However, considering the dimension is unchanged by transforming, Illingworth–Stewartson transformation is preferred in the current study.

Under the pressure gradient outside the boundary layer, the inviscid stream is assumed to satisfy the Falkner–Skan velocity distribution as

$$U_e(\bar{\xi}) = C \cdot \bar{\xi}^m, \quad (11)$$

in which C and m are constants. Parameter m denotes the direction and strength of pressure gradient; thus, it is called the pressure-gradient parameter. A positive m means a favorable pressure gradient, indicating an accelerating flow. Contrarily, a negative m means an adverse pressure gradient which decelerates a flow. The Hartree parameter $\beta = 2m/(m+1)$ is often used to symbolize the pressure gradient as well.^{28,29,35} Note that the velocity at the reference point P is prescribed; thus, the constant C can be rewritten as $C = U_{e,ref}/\bar{\xi}_{ref}^m$. The subscript “ref” denotes a quantity at the reference point. We proceed to use the Falkner–Skan transformation as

$$\eta = \bar{\eta} \sqrt{\frac{m+1}{2} \frac{U_e}{\nu_{e0} \bar{\xi}}}, \quad \xi = \bar{\xi}. \quad (12)$$

The local-similarity assumption³⁵ permits a separable-variable form for the stream function of

$$\psi = f(\eta) \sqrt{\frac{2\nu_{e0} U_e \xi}{m+1}}. \quad (13)$$

Combining Eqs. (9)–(13), one can obtain the normalized x -direction velocity as

$$f'(\eta) = \frac{U}{U_e} = \frac{u}{u_e}. \quad (14)$$

Furthermore, we supposed that the z -direction velocity and temperature are dependent solely on the local potential flow, and the normalized quantities take the forms

$$g(\eta) = \frac{w}{w_e}, \quad \tau(\eta) = \frac{T}{T_e}. \quad (15)$$

It should be noted that normalized enthalpy, instead of temperature, may also be adopted in literatures, which leads to equations differing from what we will get below. Combining the Illingworth–Stewartson transformation [Eq. (8)], the Falkner–Skan transformation [Eq. (12)], the transformed variables in Eqs. (14) and (15), and the state equation for perfect gases, one can rewrite the governing equations as

$$(Nf'')' + ff'' = \frac{2m}{(m+1)S} (f'^2 - \tau), \quad (16)$$

$$(Ng')' + fg' = 0, \quad (17)$$

$$\left(\frac{N}{Pr}\tau'\right)' + (S-1)\left(N(f'^2)'\right)' + (K-1)S(N(g^2))' + f\left(\tau' + (S-1)(f'^2)'\right) + (K-1)S(g^2)' = 0, \quad (18)$$

in which Pr is the Prandtl number equal to 0.72 throughout this paper. By applying Sutherland’s law, the Chapman–Rubesin factor N can be written as

$$N \equiv \frac{\rho\mu}{\rho_e\mu_e} = \tau^{1/2} \left(\frac{1 + T_s/T_e}{\tau + T_s/T_e} \right), \quad (19)$$

where $T_s = 110.4$ K. Ide *et al.*³⁶ adopted the assumptions of unity Prandtl number and linear law for dynamic viscosity, and the energy equation could be decoupled from the system. Parameters K and S are dependent on local variables at the boundary-layer edge. Using the isentropic relation and the flow quantities at the reference point, one can obtain

$$T_e = T_{e,ref} \left(1 + \frac{\gamma-1}{2} Ma_{e,ref}^2 \cos^2 \Lambda \right) / S, \quad (20)$$

$$K = \frac{1 + \frac{\gamma-1}{2} Ma_{e,ref}^2}{1 + \frac{\gamma-1}{2} Ma_{e,ref}^2 \cos^2 \Lambda}, \quad (21)$$

$$S = \left(1 + \frac{\gamma-1}{2} \chi^2 Ma_{e,ref}^2 \cos^2 \Lambda \right), \quad (22)$$

$$\chi = \left(\frac{\xi}{\xi_{ref}} \right)^m, \quad (23)$$

in which the specific heat ratio $\gamma = 1.4$. It can be noted that parameter S is dependent on coordinate ξ as long as the pressure gradient is not absent ($m \neq 0$), which indicates a denial of a solution with strict self-similarity. The boundary conditions are then rewritten as

$$\eta = 0 : f = f' = g = 0, \quad \tau = \tau_w \quad \text{or} \quad \tau' = 0, \quad (24)$$

$$\eta \rightarrow \infty : f' = g = \tau = 1. \quad (25)$$

So far, we have obtained transformed boundary-layer equations with an ODE form. The solution can be obtained locally by using the Runge–Kutta method. Once the boundary-layer-edge quantities at the reference point are specified, the boundary-layer profiles can be calculated locally at any location in ξ direction. Although the analytical expression of the solution cannot be obtained, such solution of an ODE system can be categorized as an exact solution.³⁷

When the Mach number approaches zero, Eqs. (16) and (17) are reduced to the equations obtained by Cooke²⁸ for incompressible boundary layers. Thus, the current boundary layer can be deemed to be a compressible form of FSC flow, or namely, compressible FSC boundary layer. Although the ODE form holds during the extension to compressible condition, the self-similarity is lost when the Mach number plays a role. Instead, Anderson³⁵ suggested local self-similarity for describing the character in such boundary layers. Due to the absence of strict self-similarity, the current boundary layers need to be calculated from location to location if the streamwise variations are interested in. Compared with the boundary-layer equations with PDE forms, the calculation in current model is largely simplified.

When the local swept angle Λ approaches zero, the z -direction velocity in potential flow vanishes and the flow in Fig. 1 is reduced to a 2D boundary layer. Under such condition, K is reduced to 1 in Eq. (21). Consequently, the energy equation [Eq. (18)] becomes independent of z -direction velocity g or w . The boundary condition in Eq. (25) can be modified as $g=0$. Combined with Eq. (17), one can obtain $w=0$. Thus, the governing equations are reduced to a 2D form correspondingly. Moreover, Eqs. (16) and (18) are reduced to equations describing a 2D compressible boundary layer with pressure gradient

and coincide with those obtained by Cohen and Reshotko²⁹ if the parameter N is set to be unity. In the absence of pressure gradient ($m = 0$), the equations are reduced to those with similar solutions presented by Schlichting and Gersten³⁴ on condition that $N = 1$ and $Pr = 1$. In this case, the boundary layer is a compressible form of Blasius boundary layer since it is reduced to the classical Blasius boundary layer when the Mach number vanishes.

Therefore, the compressible FSC boundary layer obtained in this paper can be regarded as the generalizations of the classical flows on a flat plate, namely, the incompressible FSC boundary layer and the 2D compressible Blasius boundary layer, both of which possess exact solutions with self-similarity. Based on current model, one may expect to extend any boundary-layer characteristics concerned when compressibility or crossflow plays a role.

C. Expression and solution

After solving the ODE system, the velocity needs to be expressed locally to obtain the streamwise and crosswise components. Using the local swept angle Λ , one may find that the streamwise velocity $\bar{u}$ and crosswise velocity $\bar{w}$ can be expressed as

$$\bar{u} = \frac{L^2 f' \cos^2 \Lambda + g \sin^2 \Lambda}{L^2 \cos^2 \Lambda + \sin^2 \Lambda}, \quad (26)$$

$$\bar{w} = \frac{L(g - f') \cos \Lambda \sin \Lambda}{L^2 \cos^2 \Lambda + \sin^2 \Lambda}, \quad (27)$$

in which

$$L = \chi \sqrt{\frac{\left(1 + \frac{\gamma - 1}{2} Ma_{e,ref}^2\right)}{K \cdot S}}. \quad (28)$$

According to Eq. (27), it is noted that the crossflow vanishes in two special cases.

I. $\Lambda = 0^\circ$. In this case, the z -direction velocity in the external flow vanishes and the flow is reduced to a 2D form.

II. $g = f'$. This indicates a boundary layer with zero pressure gradient, i.e., $m = 0$. Under such condition, Eqs. (16) and (17) share the same form; thus, f' and g have an identical solution by further considering the same boundary conditions. Although neither f' nor g is zero in this case, the flow is essentially two-dimensional in an oblique plane.

Special attention should be paid to the case in which $\Lambda = 90^\circ$. It can be easily verified that this condition produces zero crossflow as well. However, this condition also implies zero velocity in the x direction, i.e., $u_e = 0$, which is not allowed in current flow model according to Eqs. (12) and (14).

In boundary layers, the flow variables are usually expressed with a scaled length, which should be capable of characterizing the thin layer. We propose a length scale as follows for the compressible boundary layer with pressure gradient. Combining Eqs. (8) and (12), one can obtain

$$y = \sqrt{\frac{\nu_{e0} \xi}{U_e} \frac{a_{e0} \rho_{e0}}{a_e \rho_e}} \sqrt{\frac{2}{m+1}} \int_0^\eta \tau d\eta. \quad (29)$$

The length scale is suggested to be

$$l = \sqrt{\frac{\nu_{e0} \xi}{U_e} \frac{a_{e0} \rho_{e0}}{a_e \rho_e}}. \quad (30)$$

It is noted that l is dependent on pressure gradient, while in the limit of zero pressure gradient ($m = 0$), Eq. (30) is reduced to $\sqrt{\nu_e x / u_e}$, which is in accordance with that in 2D compressible boundary layers.³⁸

Based on Eq. (30), the wall-normal length can be scaled as

$$\bar{y} \equiv \frac{y}{l} = \sqrt{\frac{2}{m+1}} \int_0^\eta \tau d\eta. \quad (31)$$

The “bars” in Eqs. (26), (27), and (31) will be dropped in what follows.

Thus, we have obtained the normalized velocity and temperature distributions of a compressible FSC boundary layer. As a model flow, the compressible FSC boundary layer owns fewest parameters to characterize crossflow, qualitatively and quantitatively. On the one hand, crossflow is often encountered practically in the flow on a curved wall or driven by wall rotation. While in essence, it is the pressure gradient misaligned with streamwise direction that accounts for the existence of crossflow. In the present model, the pressure-gradient parameter m and local swept angle Λ act together to create the misaligned pressure gradient, and absence of either one makes it impossible. Besides, any additional effects are not introduced, thus making the model simplest. On the other hand, different distributions of crossflow can be obtained by varying the two parameters. This makes it possible to create crossflow with different directions and strengths, which serves to facilitate the fundamental research about the effects induced by crossflow.

Gaponov and Smorodskii³² and Tempelmann *et al.*²⁴ used similar equation systems to investigate the stability of compressible boundary layers with crossflow. As a matter of fact, besides the prescribed variables f , g , and Θ in the two papers, the density ρ and the Chapman–Rubesin factor N (denoted as C therein) are unknown variables as well. As the unknown variables outnumber the equations, the system cannot be considered to be well-posed. While in the present study, the normalized temperature is adopted instead of the enthalpy. This option turns the two unknown variables to be functions of temperature, which makes the system well-posed. Furthermore, in order to solve the equations, local quantities at boundary layer edge need to be specified previously at each location. For this reason, Gaponov and Smorodskii³² applied a quite rough approximation of self-similarity to obtain the solution, and Tempelmann *et al.*²⁴ adopted the solution as upstream boundary condition for the PDE-form boundary-layer equations. In comparison, the equation system present in the current paper provides a simpler way to resolve the 3D boundary layer with appropriate assumptions.

D. Skin friction and heat transfer

The skin friction and heat transfer at the wall are the primary interest in the study of compressible boundary layers. In the coordinate system (x, y, z) , the local skin friction components can be obtained as

$$\theta_{w,x} \equiv \left(\mu \frac{du}{dy} \right)_w = \mu_w \rho_w u_e \sqrt{\frac{m+1}{2}} \sqrt{\frac{u_e}{\rho_e \mu_e x}} f_w'', \quad (32)$$

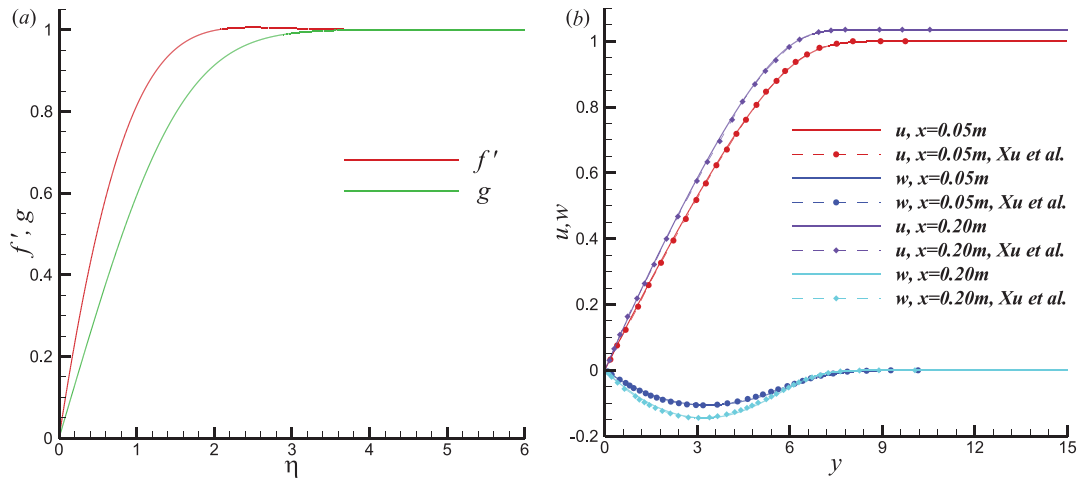


FIG. 2. Typical profiles of velocities ($Ma = 3.8$, $T_e = 120$ K, $T_w = T_{ad}$). (a) Normalized velocities in x direction [f'] and z direction [g]. (b) Streamwise velocity [u] and crosswise velocity [w]. The velocity components at $x = 0.20$ m are scaled by the velocity value at $x = 0.05$ m.

$$\theta_{w,z} \equiv \left(\mu \frac{dw}{dy} \right)_w = \mu_w \rho_w w_e \sqrt{\frac{m+1}{2}} \sqrt{\frac{u_e}{\rho_e \mu_e x}} g'_w. \quad (33)$$

Introduce the local swept angle Λ_1 at the location concerned, and it satisfies

$$\cos \Lambda_1 = \frac{L \cos \Lambda}{\sqrt{L^2 \cos^2 \Lambda + \sin^2 \Lambda}}, \quad \sin \Lambda_1 = \frac{\sin \Lambda}{\sqrt{L^2 \cos^2 \Lambda + \sin^2 \Lambda}}. \quad (34)$$

The components of skin-friction coefficient related to local dynamic pressure of the external flow are

$$C_{f,x} \equiv \frac{\theta_{w,x}}{\frac{1}{2} \rho_e u_{e, \text{str}}^2} = N_w (\cos \Lambda_1)^{3/2} \sqrt{2(m+1)} \frac{1}{\sqrt{Re}} f''_w, \quad (35)$$

$$C_{f,z} \equiv \frac{\theta_{w,z}}{\frac{1}{2} \rho_e u_{e, \text{str}}^2} = N_w \sqrt{\cos \Lambda_1} \sin \Lambda_1 \sqrt{2(m+1)} \frac{1}{\sqrt{Re}} g'_w, \quad (36)$$

in which N is the Chapman–Rubesin factor, and $Re = \frac{\rho_e u_{e, \text{str}} x}{\mu_e}$. The subscript *str* denotes the streamwise direction. Thus, the total skin-friction coefficient can be obtained as

$$C_f = N_w \sqrt{2(m+1)} \frac{\sqrt{\cos \Lambda_1}}{\sqrt{Re}} \sqrt{(f''_w \cos \Lambda_1)^2 + (g'_w \sin \Lambda_1)^2}. \quad (37)$$

When $\Lambda = 0^\circ$ and $m = 0$, the skin friction in Eq. (37) is reduced to the 2D form.³⁵ In comparison, the occurrence of crossflow alters both the direction and magnitude of skin friction.

In a stream-fitted coordinate system, the components of skin-friction coefficient are

$$C_{f, \text{str}} = N_w \sqrt{2(m+1)} \frac{\sqrt{\cos \Lambda_1}}{\sqrt{Re}} (f''_w \cos^2 \Lambda_1 + g'_w \sin^2 \Lambda_1), \quad (38)$$

$$C_{f, \text{cr}} = N_w \sqrt{2(m+1)} \frac{(\cos \Lambda_1)^{3/2} \sin \Lambda_1}{\sqrt{Re}} (g'_w - f''_w), \quad (39)$$

in which the subscript *cr* denotes the crosswise direction. It can be verified that Eqs. (38) and (39) can produce the same expression of total skin-friction coefficient as shown in Eq. (37).

In practice, the boundary layer is attached to the wall which is usually not adiabatic. Thus, the heat transfer at the wall is of great concern especially in supersonic and hypersonic conditions. According to the Fourier's law of heat conduction, the heat transfer at the wall can be obtained as

$$q_w \equiv \left(\kappa \frac{\partial T}{\partial y} \right)_w = \kappa_w \sqrt{\frac{m+1}{2}} \sqrt{\frac{u_{e, \text{str}}}{\rho_e \mu_e x}} \rho_w T_e \tau'_w. \quad (40)$$

Then the local Stanton number can be further obtained as

$$St \equiv \frac{q_w}{\rho_e u_{e, \text{str}} c_p (T_{ad} - T_w)} = \frac{\kappa_w \rho_w}{\kappa_e \rho_e Pr} \frac{1}{\sqrt{Re}} \sqrt{\frac{m+1}{2}} \frac{\tau'_w}{(\tau_{ad} - \tau_w)}, \quad (41)$$

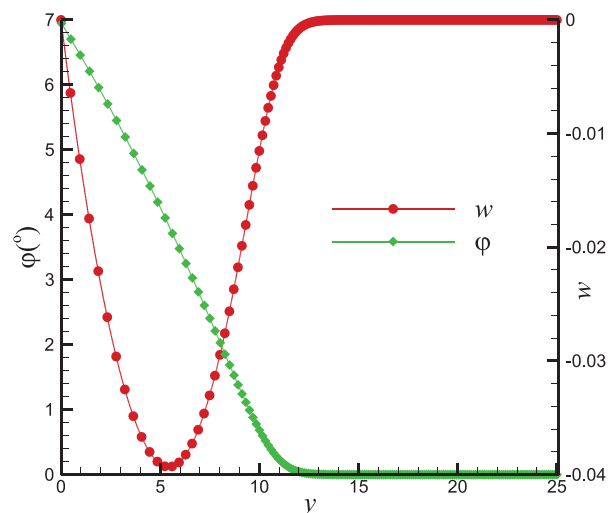


FIG. 3. Distributions of crossflow and deviation angle ($Ma = 4.5$, $T_e = 65.15$ K, $m = 0.1$, $\Lambda = 30^\circ$, $T_w = T_{ad}$).

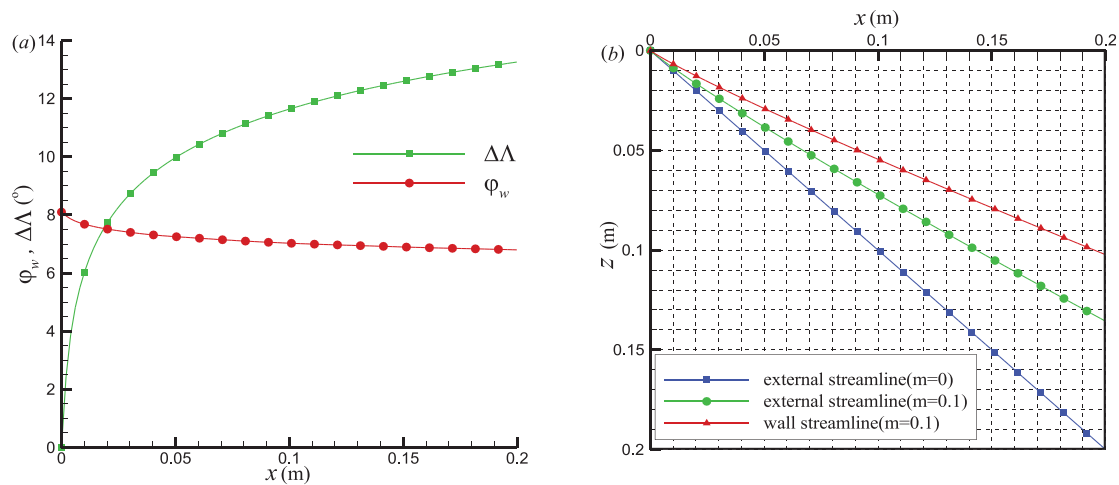


FIG. 4. Deviations of external streamline and wall streamline ($Ma = 0.5$, $T_w = T_{ad}$, $m = 0.1$, $\Lambda = 45^\circ$, $T_w = T_{ad}$). (a) Variations of maximum deviation angle [ϕ_w] and local-swept-angle difference [$\Delta\Lambda$]. (b) External streamlines and wall streamline.

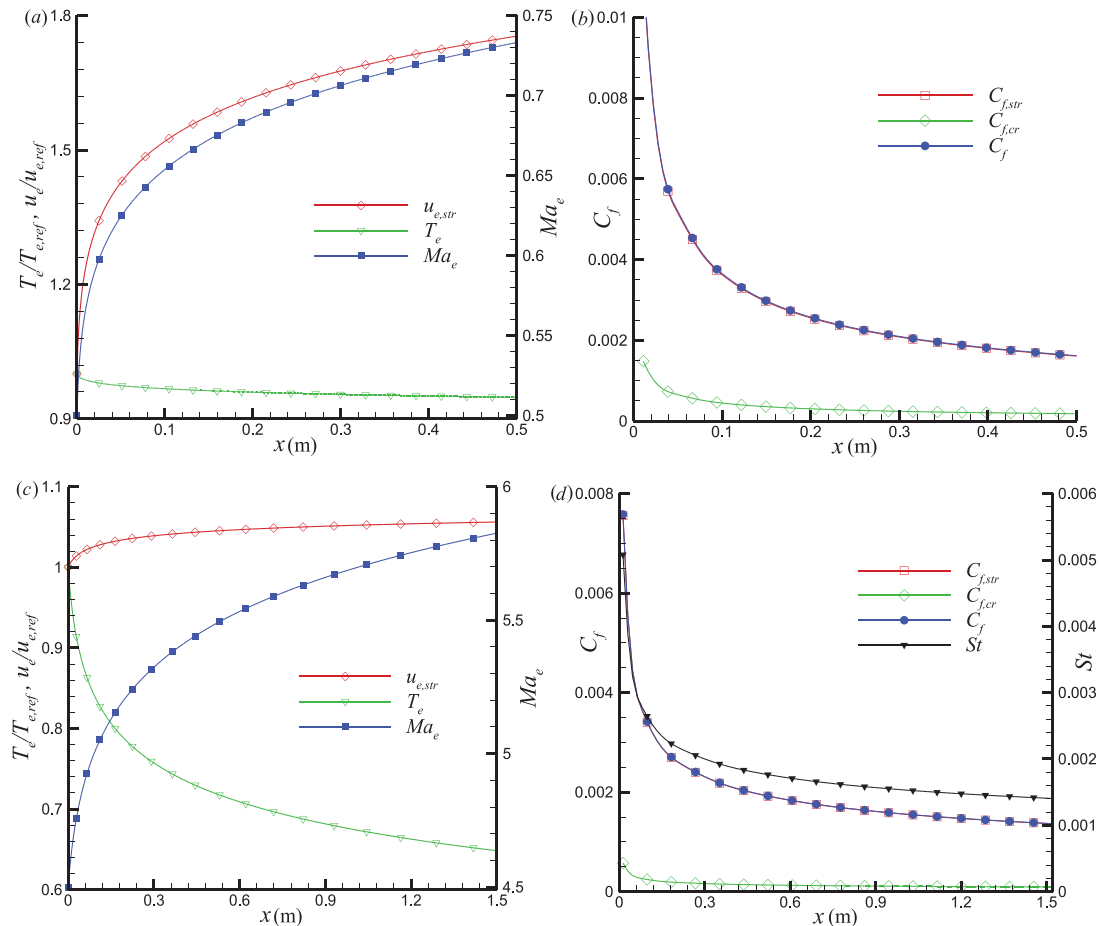


FIG. 5. Streamwise variation of boundary-layer characteristics. Subsonic case: $Ma = 0.5$, $T_w = T_{adi}$; supersonic case: $Ma = 4.5$, constant wall-temperature, $T_{w,ref} = 2.0$. Other parameters can be referred to Figs. 4 and 3. (a) External-flow quantities [subsonic]. (b) Skin-friction coefficient [subsonic]. (c) External-flow quantities [supersonic]. (d) Skin-friction coefficient and Stanton number [supersonic].

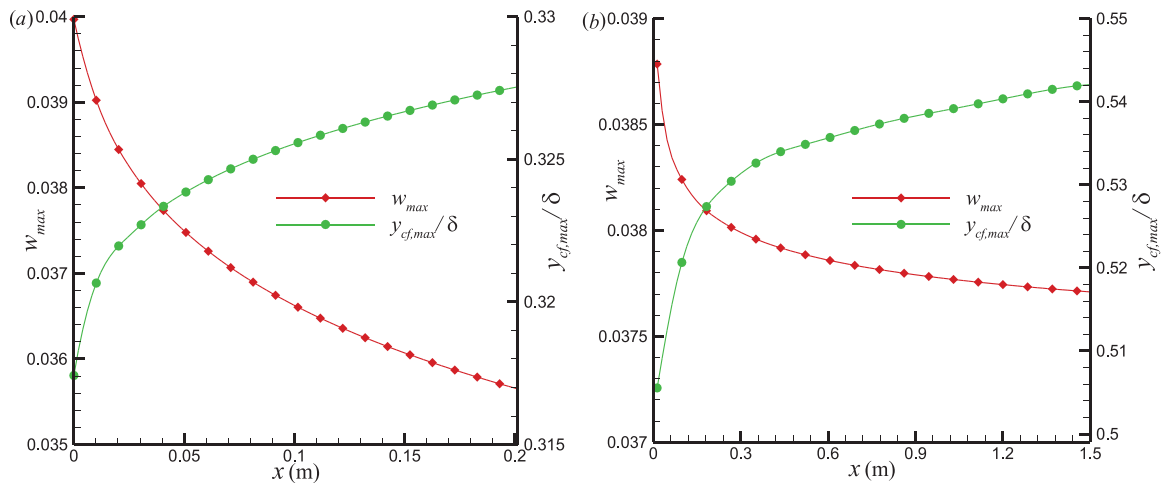


FIG. 6. Magnitude of peak crossflow and the location in boundary layer. (a) Subsonic case as in Fig. 4. (b) Supersonic case as in Fig. 3. Subscript *cf* denotes crossflow. The boundary layer thickness satisfies $u(\delta) = 0.99$.

in which the subscript *ad* denotes an adiabatic wall. It can be found out from Eqs. (37) and (41) that a larger m produces larger skin friction and heat transfer. It is in good agreement with the common sense that a more accelerating boundary layer is subject to more severe drag force and heat conduction than a less accelerating or a decelerating flow.

In some practical cases, the skin friction and heat transfer scaled uniformly by certain quantities are more concerned. The flow quantities of incoming flow, or at the reference point in current model, may serve to scale. Thus, Eqs. (35)–(41) need to be modified by factors related to $u_{e,str}/u_{e,str,ref}$.

III. RESULTS AND DISCUSSIONS

A. Properties of solutions

In order to express the flow quantities more directly, the normalized temperature is denoted as T in what follows, instead of τ . The

local features of the compressible FSC boundary layer are first considered at the reference point. The solutions of the ODE system are presented in Fig. 2 for a supersonic boundary layer. Typical distributions of velocities are plotted for an adiabatic wall with conditions considered by Xu *et al.*⁹ The x - and z -direction velocities are solved directly, and the profiles seem similar [Fig. 2(a)]. While in the stream-fitted coordinate system, the crosswise velocity component reaches a maximum in the boundary layer [Fig. 2(b)]. The classic C-shape distribution resembles the profiles on the swept configurations.¹ For comparison, we also present the results obtained by Xu *et al.*⁹ from solving the PDE system, i.e., Eqs. (1)–(4). The solution of the ODE system is revealed to be in good agreement with that of the PDE system. This comparison may serve to validate the approach presented in current paper. It is also noted that the crossflow is negative in sign under a favorable pressure gradient; and it will be seen that the reverse is

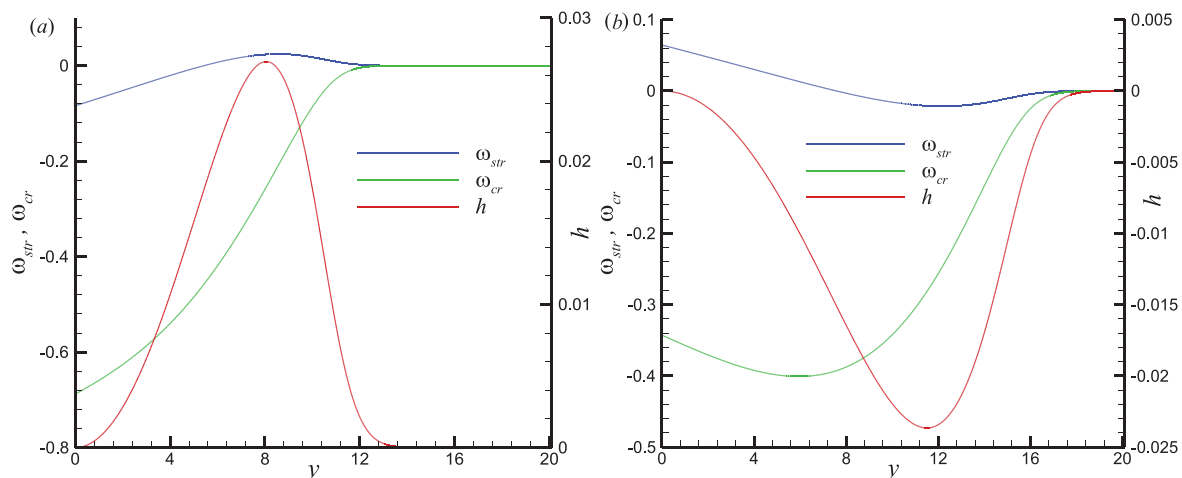


FIG. 7. Profiles of vorticity and helicity ($Ma = 4.5$, $T_0 = 65.15$ K, $\Lambda = 30^\circ$, $T_w = T_{ad}$). (a) $m = 0.1$ and (b) $m = -0.05$. Vorticity $\vec{\omega}$ is defined as $\vec{\omega} \equiv \nabla \times \vec{V}$. Helicity h is defined as $h \equiv \vec{V} \cdot \vec{\omega}$.

true. Based on the sketch in Fig. 1, it can be summarized that the crossflow is always pointing to the curvature center of external streamline.

As mentioned above, the crossflow may cause skin friction to deviate from locally streamwise direction. The deviation angle φ is defined as the angle between velocity and external-flow direction. The distribution of deviation angle is plotted for the same case in Fig. 3 along with the crosswise velocity. Although the cross-flow peaks at about $y=5.5$ inside the boundary layer, the deviation angle reaches maximum at the wall and decreases outward monotonically. The maximum deviation angle is around 7° , which also quantifies the deviation of local skin friction. Based on the direction of crossflow, it can be inferred that the skin friction is deviated slightly toward the concave side of local streamline.

To reveal the discernible deviation of skin friction, a subsonic case is considered. The streamwise variation of maximum deviation

angle, denoted as φ_w , is plotted in Fig. 4(a). Note that φ_w also characterizes the deflection angle of skin friction from the direction of external flow. With the downstream development of boundary layer, the maximum deviation angle decreases by about 1° in 0.2 m. Note that the location where $x=0$ should be recognized as the reference point. In addition, the difference of local swept angle $\Delta\Lambda$, which is defined as $\Lambda-\Lambda_1$, is also plotted to exhibit the deviation of external flow from the initial direction at the reference point. A sharp increase is first observed for $\Delta\Lambda$ near the reference point, followed by a slower accretion further downstream. Based on the deviation angles, the streamlines are plotted in x - z plane [Fig. 4(b)]. For comparison, we also plot the external streamline with local swept angle unchanged under zero pressure gradient. Owing to the existence of lateral component of pressure gradient, the external flow deviates from the initial direction. Similarly, the wall streamline (streamline approaching the wall), also

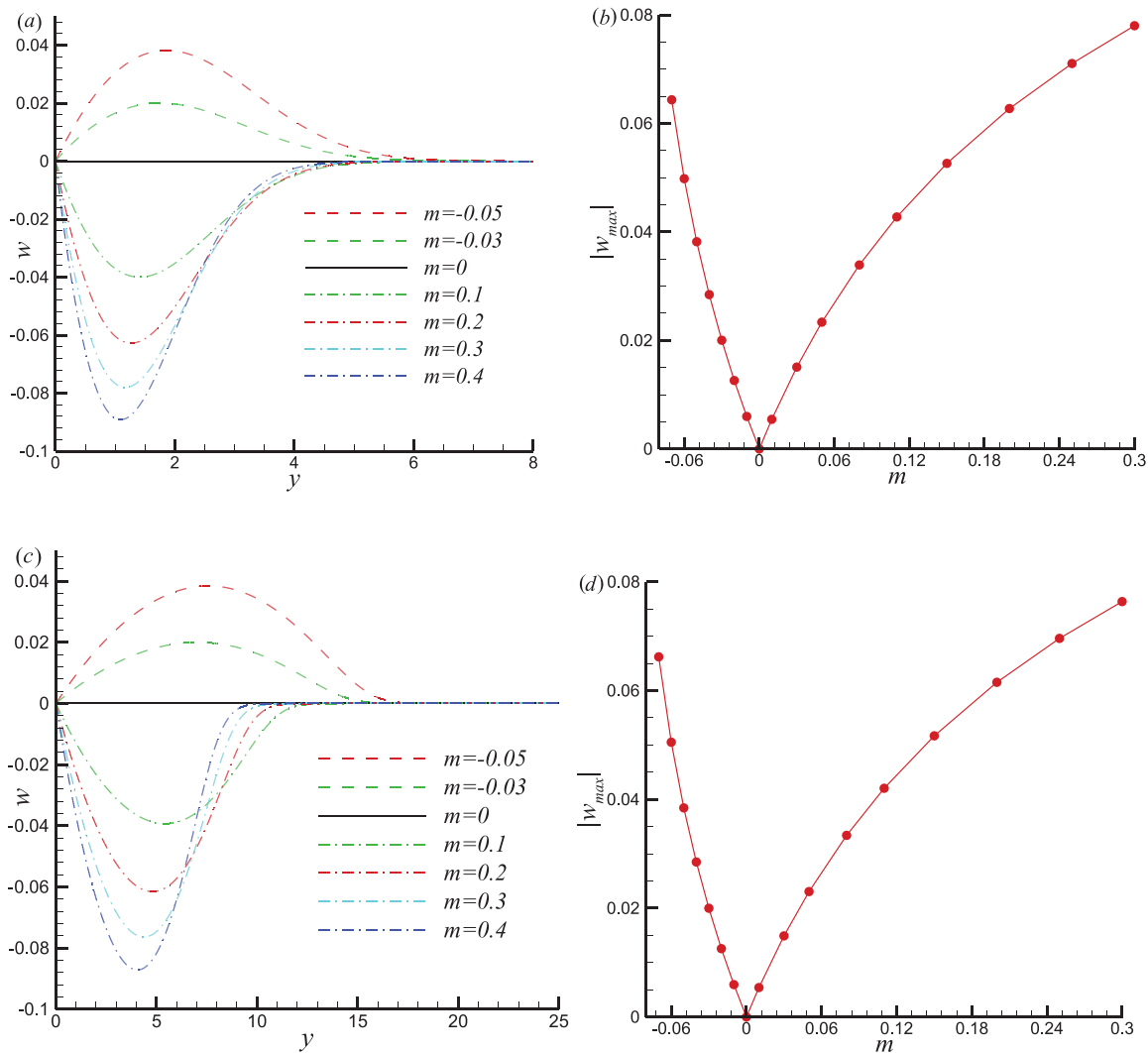


FIG. 8. Crossflow under various pressure gradients. (a) Profiles at $Ma=0.5$, $\Lambda=45^\circ$. (b) Cross-flow maxima at $Ma=0.5$, $\Lambda=45^\circ$. (c) Profiles at $Ma=4.5$, $\Lambda=30^\circ$. (d) Cross-flow maxima at $Ma=4.5$, $\Lambda=30^\circ$.

known as skin friction line, skews from the external-flow direction as a result of the existence of crossflow. The increase in $\Delta\Lambda$ also indicates a decrease in local swept angle Λ_1 downstream. This result weakens the lateral pressure gradient, which can be denoted by $m \sin \Lambda_1$. Such effect in return alleviates the deviating trend of external streamline, revealing itself as a slower increase in $\Delta\Lambda$. Besides, special attention should be paid to the streamlines with pressure gradient, which are not straight lines. Instead, the two lines are actually slightly curved with barely discernible curvature. This can be testified by the non-constant variation of deviation angles in Fig. 4(a).

Besides the direction of skin friction, the magnitude is also presented for subsonic and supersonic cases (Fig. 5). The streamwise distributions of external-flow quantities are first given in Figs. 5(a) and 5(c). The temperature and streamwise velocity are scaled by the counterparts at the reference point. Under a favorable pressure gradient, the flow is accelerated in the x direction, which results in the increase in total velocity and decrease in temperature under the isentropic assumption. The combination of both effects increases the Mach number downstream. Skin coefficient is calculated through Eqs. (37)–(39) and displayed in Figs. 5(b) and 5(d). The main feature is the unanimous decrease as a result of the dominant effect of Reynolds number, which varies with x as $Re \sim x^{-1/2}$. The crosswise component of skin-friction coefficient is too small to make any discernible contribution to the total value. Compared with the results in Fig. 4, the crossflow seems to affect the skin friction mainly on the direction rather than the magnitude. The Stanton number is calculated as well for a supersonic case with constant-temperature wall. Similar variation can be observed as the skin-friction coefficient.

As crossflow is accountable for the so-called cross-flow instability and transition, we also investigate the basic features of crossflow. The maximum is believed to be the most important characteristic related to crossflow; thus, we plot the magnitude and location inside the boundary layer (Fig. 6). Both cases reveal a streamwise decrease in maximum crossflow, denoted as w_{max} . This variation may be ascribed to the decrease in lateral pressure gradient as discussed previously. The location where the crossflow reaches its maximum is denoted as

$y_{cf,max}$ the height above the wall. In Fig. 6, the height is scaled by the local boundary-layer thickness. Although the scaled height varies in different ranges for the subsonic and supersonic cases, the peak of crossflow lies mainly in the lower half of boundary layer, and the monotonic increase is observed in both cases. As the boundary layer grows downstream, it can be inferred that the height grows faster than boundary layer itself does.

In comparison with the 2D boundary layers, the emergence of crossflow alters the distribution of vorticity by creating a streamwise component. The typical profile of vorticity in a 3D boundary layer is shown in Fig. 7. A non-zero distribution of streamwise vorticity is observed inside the boundary layer. Sign change happens at the location where crossflow reaches its maximum. Although the streamwise component is much weaker than the crosswise component, it may serve as a source of streamwise vortex during the process of crossflow-dominated boundary-layer transition. Helicity appears particularly in 3D boundary layers as a result of the occurrence of streamwise component of vorticity, and it was introduced as an indicator of cross-flow strength in the prediction of cross-flow transition.³⁹ Based on the current model, the distribution of helicity is presented. The value is non-negative under a favorable pressure gradient [Fig. 7(a)], while non-positive under an adverse one [Fig. 7(b)]. Similar to the profile of crossflow, helicity has zero values at two ends and reaches a maximum at the upper half of boundary layer.

Basic features of the compressible FSC boundary layer are investigated. It is revealed that the current model is capable of creating crossflow with classical C-shape velocity profile. The result shows that the crossflow is always pointing to the curvature center, regardless of favorable or adverse pressure gradients. The crossflow reaches its maximum at the lower half of boundary layer with magnitude of several percentage of external flow. The existence of crossflow distorts the boundary layer and produces a crosswise component of skin friction. Although the crosswise skin friction makes quite limited contribution to the total value, it results in noticeable deviation of wall streamline from the locally streamwise direction. With the downstream increase in the deviation angle, it is revealed that the external flow becomes

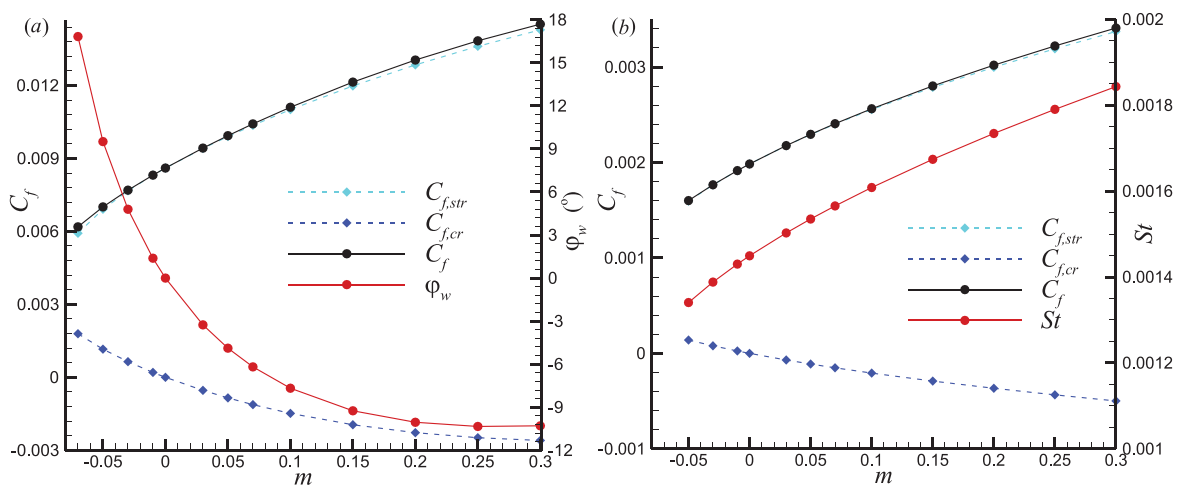


FIG. 9. Variations of skin-friction coefficient and Stanton number against pressure-gradient parameter. (a) $Ma = 0.5$, $T_w = T_{ad}$. (b) $Ma = 4.5$, $T_w = 2.0$.

more and more aligned with the direction of pressure gradient, which decreases the lateral pressure gradient and consequently weakens the cross-flow effect. The occurrence of crossflow in 3D boundary layers also distorts the distribution of vorticity by producing a streamwise component. In addition, helicity appears inside the boundary layer with a profile similar to the crossflow.

B. Effects of flow parameters

According to the present flow model, the crossflow is created in the presence of pressure gradient and local swept angle. Thus, it can be suggested as the primary importance of these two factors. Previous results uncover some combined effects of parameter m and Λ along the streamwise direction. More details of cross-flow characteristics under the effects of flow parameters are studied in this sub-section. For simplicity, the boundary layer just at the reference point is considered.

1. Pressure gradient

The subsonic and supersonic cases are investigated again at fixed local swept angles (Fig. 8). The behavior of crossflow in two cases seems similar, and it suggests that the magnitude of crossflow is positive under adverse pressure gradients and negative under favorable pressure gradients. The variations of cross-flow maximum in Figs. 8(b) and 8(d) reveal the increasing strength of crossflow with parameter m . This result signifies a larger pressure gradient usually produces a stronger crossflow. A closer look at Figs. 8(a) and 8(c) suggests that the boundary layer under an adverse pressure gradient is commonly thicker than that in a favorable one. Furthermore, more favorable pressure gradients seem to produce thinner boundary layers, and more adverse pressure gradients produce thicker ones. The results reveal continuous and monotonic variation of crossflow under varying pressure gradients, which suggests that m can be an effective parameter to control crossflow.

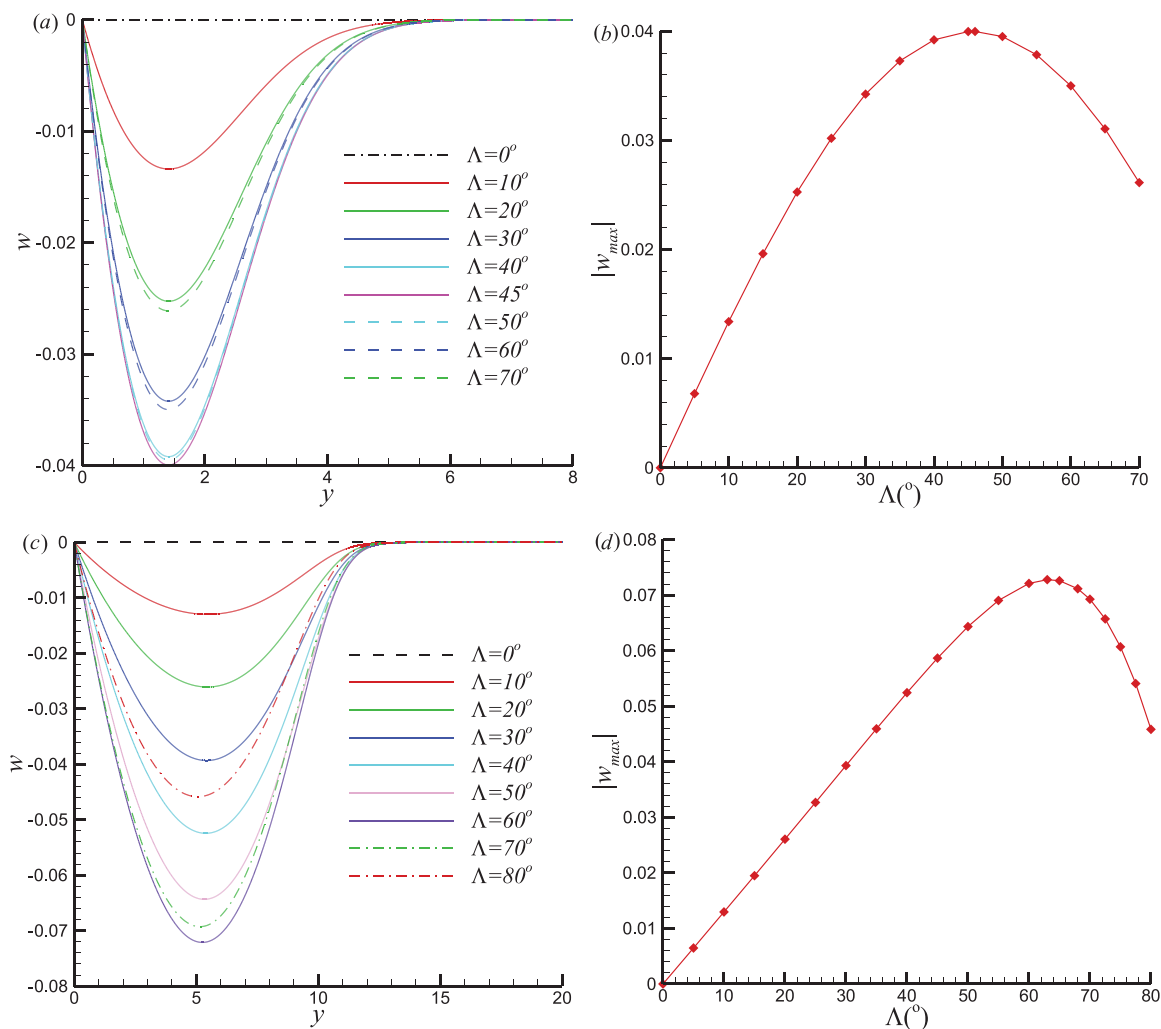


FIG. 10. Crossflow under various local swept angles ($m = 0.1$). (a) Profiles at $Ma = 0.5$, (b) cross-flow maxima at $Ma = 0.5$, (c) profiles at $Ma = 4.5$, and (d) cross-flow maxima at $Ma = 4.5$.

The effects of pressure gradient on skin friction and heat transfer at the wall are studied and shown in Fig. 9. The streamwise component of skin-friction coefficient is found to increase with pressure-gradient parameter m , while the crosswise component varies contrarily. The total skin-friction coefficient is increasing due to the major contribution of streamwise component. Considering the opposite directions for negative and positive m , the crosswise skin-friction coefficient indeed increases in value with increasing strength of pressure gradient, so does the skin-friction deviation angle φ_w . The behaviors of $C_{f,cr}$ and φ_w are ascribed to the variation of crossflow itself. Besides, for a supersonic boundary layer on a cooled wall, the Stanton number is found to increase with m , which implies that accelerating a flow enhances the heat transfer at the wall. Furthermore, the variations of skin friction and heat transfer are in good agreement with the results in Fig. 5.

Generally, a stronger pressure gradient, no matter a favorable one or an adverse one, produces a stronger crossflow. In the current flow

model, the variation of crossflow seems to be more sensitive to m with negative values (i.e., adverse pressure gradients). For the same absolute value of m , an adverse pressure gradient produces stronger crossflow and thicker boundary layer than a favorable one.

2. Local swept angle

The cross-flow velocity profiles and maxima under varying local swept angle are presented in Fig. 10. In comparison with the effects of pressure gradient, the most striking difference must be the non-monotonic variation of crossflow with the increase in local swept angle. There exists a peak of the maximum curve for a fixed pressure-gradient parameter m . Denote w_p as the peak value of maximum crossflow when local swept angle varies and Λ_p as the corresponding swept angle. The cross-flow velocity increases from zero when Λ increases in the range $\Lambda < \Lambda_p$ and decreases in the range $\Lambda > \Lambda_p$. For the subsonic and

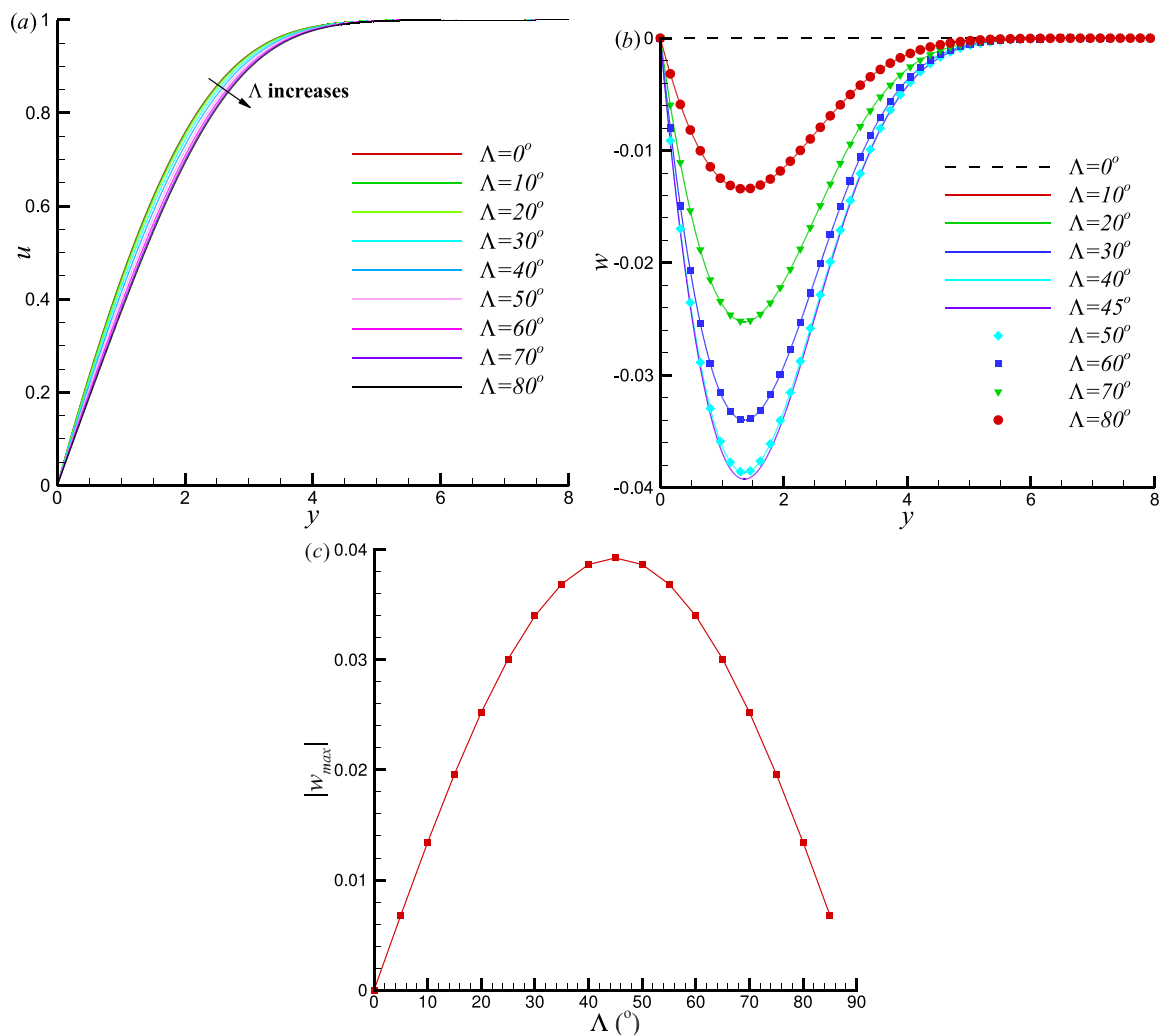


FIG. 11. Velocity profiles and cross-flow maxima of incompressible FSC boundary layer under various local swept angles ($m = 0.1$). (a) Streamwise velocity profiles, (b) cross-flow velocity profiles, and (c) cross-flow maxima.

supersonic cases, Λ_p is about 46° and 63° , respectively. Besides, varying local swept angle barely alters the boundary-layer thickness, which differs from the effect of pressure gradient as well.

For the classical FSC flow, the solution exhibits some particularities. Considering the flow model by setting $Ma = 0$, the local swept angle Λ vanishes in the governing equations [Eqs. (16) and (17)] and boundary conditions [Eqs. (24) and (25)]. Thus, the solution f' and g become independent of Λ . Under such condition, the local swept angle affects crossflow through Eq. (27). At zero Mach number, Eq. (27) is reduced to

$$w = (g - f') \cos \Lambda \sin \Lambda, \quad (42)$$

which implies that the crossflow at angle Λ is identical to that at angle $90^\circ - \Lambda$. Results in Fig. 11 confirm this point clearly. The cross-flow velocity profiles coincide for complementary swept angles as shown in Fig. 11(b). Equation (42) also indicates that the crossflow is largest when $\Lambda = 45^\circ$, as displayed in Fig. 11(c). Nevertheless, the streamwise velocity [Eq. (26)] does not share the same particularity; thus, a monotonic variation is observed when Λ increases [Fig. 11(a)]. When the Mach number deviates from zero, the compressibility effect plays a role and the crossflow profiles for complementary swept angles deviate from each other as well [Figs. 10(a) and 10(c)]. Correspondingly, Λ_p gradually becomes larger than 45° .

It has been found that the peak value of cross-flow maximum is reached at different swept angles for different pressure gradients. Extensive calculations provide more information about the peak value, as displayed in Fig. 12. Generally, the variation of peak value against pressure-gradient parameter resembles that of cross-flow maximum in Fig. 8. This can be regarded as the general effect of pressure gradient on the strength of crossflow. As for the local swept angle, Λ_p decreases monotonically with the increase in m from negative to positive values. For the subsonic case, the variation for Λ_p is within 0.4° around 46° , while for the supersonic case, the variation is about 10° in almost the same range of m . The result indicates that Λ_p is more sensitive to pressure gradient in a more compressible boundary layer.

Variations of skin-friction coefficient and Stanton number at varying local swept angle are presented in Fig. 13. Similar to the variation of cross-flow maximum as shown in Fig. 10(d), the crosswise component of skin-friction coefficient owns a peak; however, the peak locates at $\Lambda \approx 55^\circ$, differing from Λ_p . The shift of the peak's location can be explained by considering Eq. (39), in which the crosswise component is controlled by the crossflow [Fig. 10(c)] together with the factor $\sqrt{\cos \Lambda_1}$ (reduced to $\sqrt{\cos \Lambda}$ at the reference point). It is this factor that accounts for the peak shift. The variation of streamwise component can also be explained by the influence of this factor. The total skin-friction coefficient decreases monotonically with the swept angle due again to the dominant contribution of the streamwise component. In comparison, the Stanton number increases slightly with the swept angle. Analysis of Eq. (41) reveals that the variation is ascribed mainly to the increase in temperature gradient at the wall, as displayed in Fig. 13(c). Another reason is the fairly slight variation of adiabatic-wall temperature in the whole range of swept angle. In an even more compressible case, T_{ad} varies less than 2% for adverse and favorable pressure gradients (Fig. 14).

Under the effect of local swept angle, the non-monotonic variation of crossflow exists as a fundamental feature. It has been shown that the strength of crossflow peaks at a specific swept angle no less than 45° . The deviation angle and crosswise skin friction exhibit similar variations. Compared with the temperature, velocity is found to be more sensitive to the variation of local swept angle.

3. Mach number

One success of current flow model is extending the classical FSC flow to compressible conditions. In a practical case, the 3D boundary layer at some specific Mach number is mainly concerned. However, the general properties related to crossflow at varying Mach number are essential to be prior and fundamental knowledge. The cross-flow profiles are plotted for different Mach numbers under the same pressure-gradient parameter, local swept angle, and wall condition (Fig. 15). The strength of crossflow seems to increase with Mach

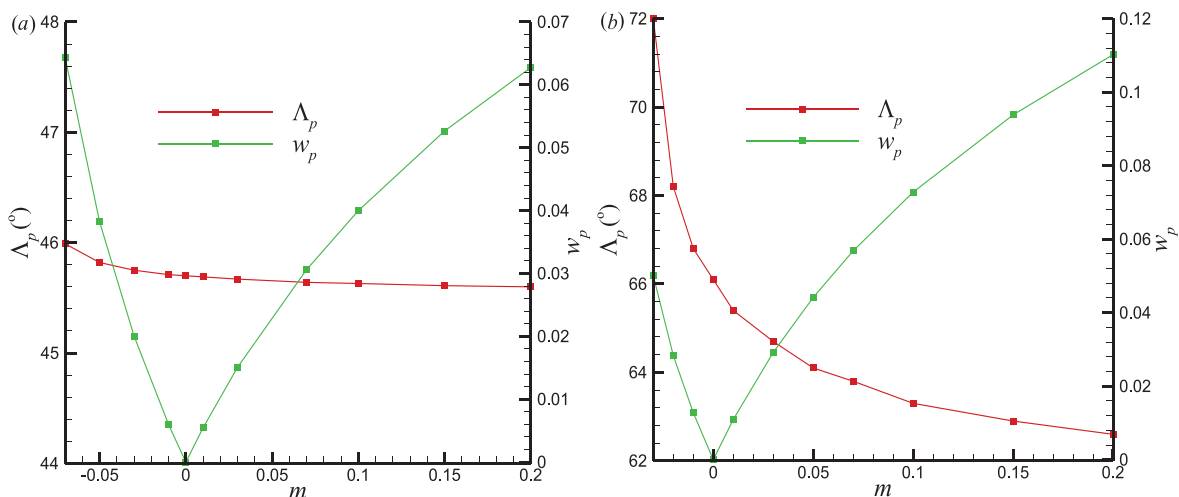


FIG. 12. Peak value of cross-flow maximum and corresponding swept angle at varying pressure gradient. (a) $Ma = 0.5$ and (b) $Ma = 4.5$. Note that Λ_p can be any value at $m = 0$, and a mean value is chosen to produce a continuous variation.

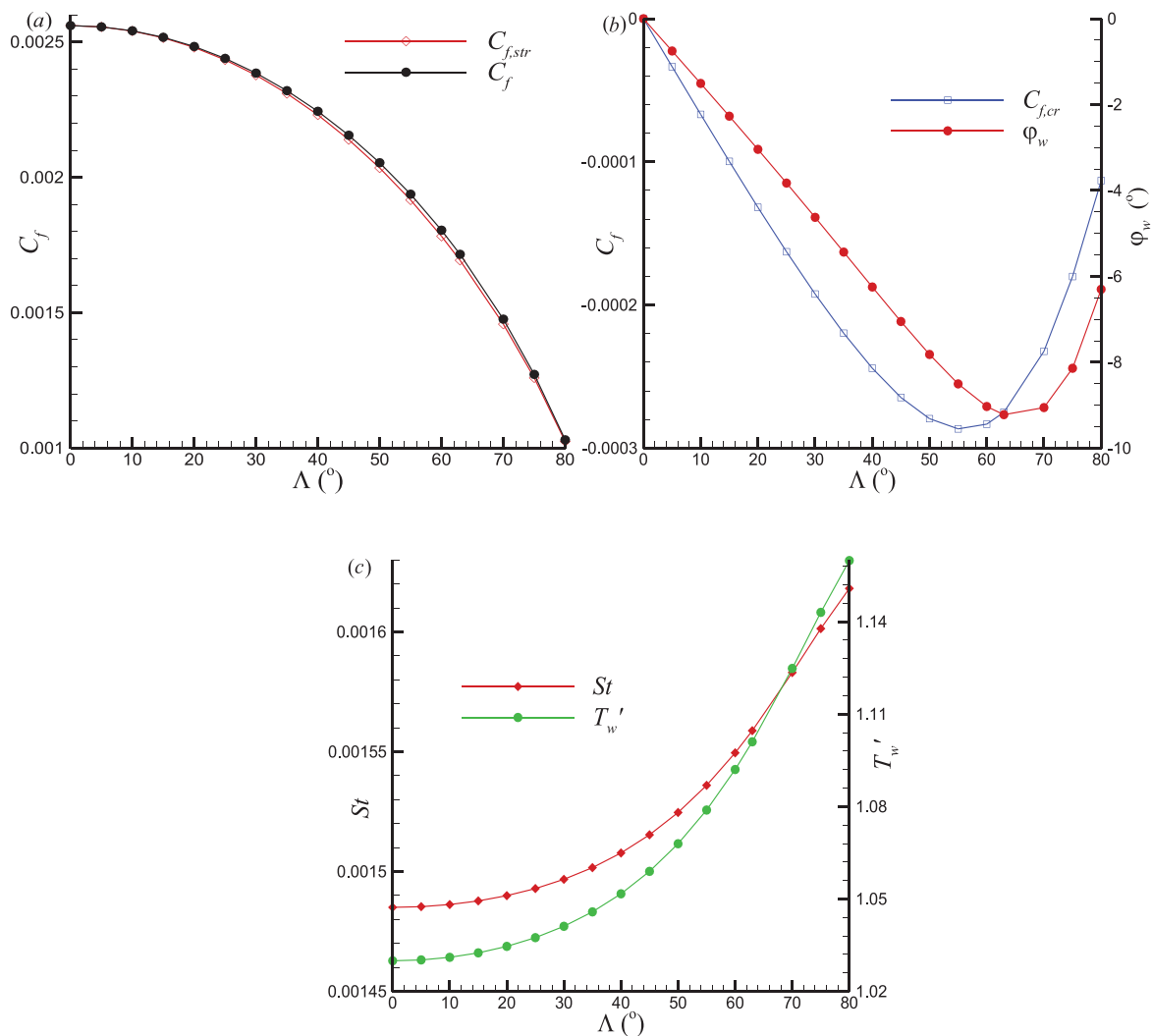


FIG. 13. Variations of skin-friction coefficient and Stanton number against local swept angle ($Ma = 4.5$, $T_w = 2.0$). (a) Streamwise and total skin-friction coefficient, (b) cross-wise skin-friction coefficient, and (c) Stanton number and temperature gradient at the wall.

number, and the boundary layer is observed to get thicker at higher Mach numbers. Considering the length scale presented in Eq. (30), the difference of thickness is not as large as it looks in the figure.

The results in Fig. 12 have indicated the increases of cross-flow peak w_p and the corresponding swept angle Λ_p at increasing Mach numbers. More detailed calculations have been conducted to reveal the variations as shown in Fig. 16. Monotonic increases of Λ_p and w_p are observed for both adverse and favorable pressure gradients. In comparison, the cross-flow peak under an adverse pressure gradient exhibits a sharper increase than under a favorable one. So does the swept angle Λ_p . Although a minor difference is revealed for the swept angle between favorable pressure gradients, the cross-flow peak differs much in the whole range of Mach number. The trend of increase in Fig. 16(b) seems to be inconsistent with the slow increase in cross-flow maximum in Fig. 15 at high Mach numbers. This phenomenon results from the prescribed swept angle (45°) in Fig. 15, which deviates

further from the increasing Λ_p at increasing Mach numbers. A direct consequence is that the cross-flow maximum deviates further from the peak value as displayed in Fig. 16(b); thus, the increase seems to be slowed.

Thermal effect is another feature of compressible boundary layer due to the coupling of velocity and temperature in the governing equations. The adiabatic wall temperature on an insulated wall affects the heat transfer in direction and magnitude, as indicated in Eq. (41). As an ingredient of the solution, the adiabatic wall temperature is calculated in 3D compressible boundary layers at different Mach numbers (Fig. 17). An adverse pressure gradient tends to produce a slightly higher adiabatic wall temperature, and the reverse is true for a favorable pressure gradient. In general comparison with the 2D boundary layer, however, the crossflow causes quite limited variation of adiabatic wall temperature. It is revealed again that the thermal property is less sensitive to the cross-flow effect.

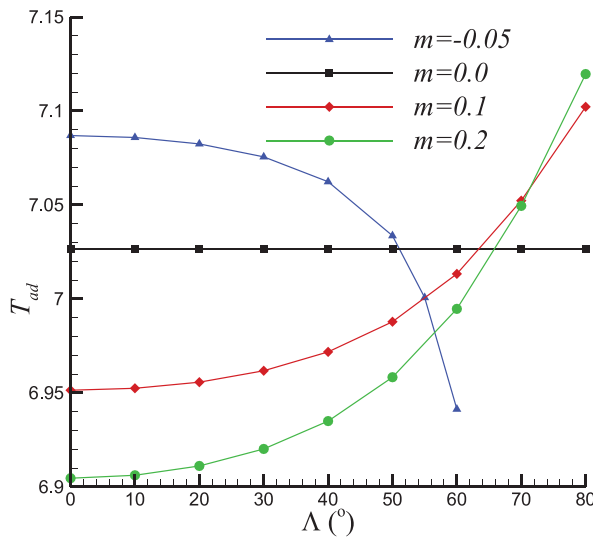


FIG. 14. Variation of adiabatic wall temperature against local swept angle ($Ma = 6.0$).

4. Wall temperature

The temperature on a non-adiabatic wall affects not only the heat transfer but also the velocity profiles inside the boundary layer. The effect of wall temperature on the crossflow is studied on cooled and heated walls. It is revealed that a warmer wall tends to create a stronger crossflow, as shown in Fig. 18. By looking carefully at the curves near $y = 0$, one may observe a contrary variation for the slope of the curves. Thus, a decrease in crosswise skin friction could be expected. However, it should be borne in mind that the wall-normal coordinate y is dependent on the temperature distribution as indicated by Eq. (31), and temperature distribution is heavily determined by wall temperature. Therefore, the profiles are stretched by different extents for

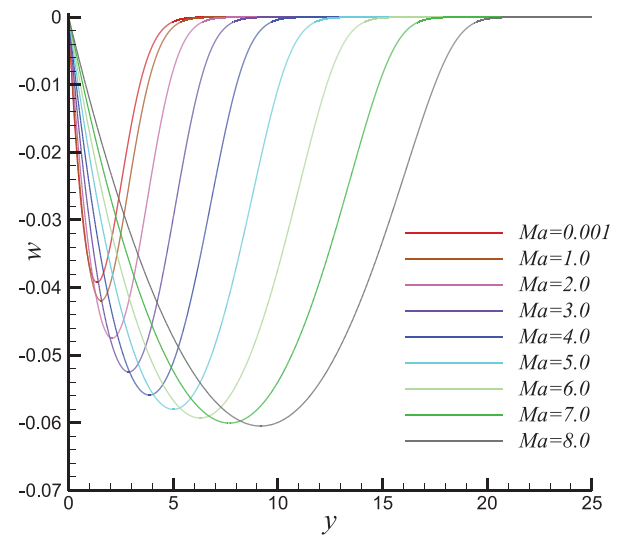


FIG. 15. Cross-flow velocity profiles at various Mach numbers ($m = 0.1$, $\Lambda = 45^\circ$, $T_w = T_{ad}$).

different cases in Fig. 18. The variation of skin-friction coefficient is calculated against wall temperature based on Eqs. (37)–(39) and displayed in Fig. 19. The crosswise component increases in magnitude with wall temperature although much smaller than the streamwise one. The variation of crosswise skin friction is consistent again with that of crossflow strength.

The enhancing effect of warmer wall on the crossflow is quantified by considering the variation of cross-flow peak, which exhibits monotonic increase with wall temperature (Fig. 20). In the temperature range considered, the magnitude of cross-flow peak increases over 100%, while the corresponding swept angle decreases no more than 2° (around 3%). The comparison indicates that wall temperature affects the crossflow mainly in the magnitude rather than the direction.

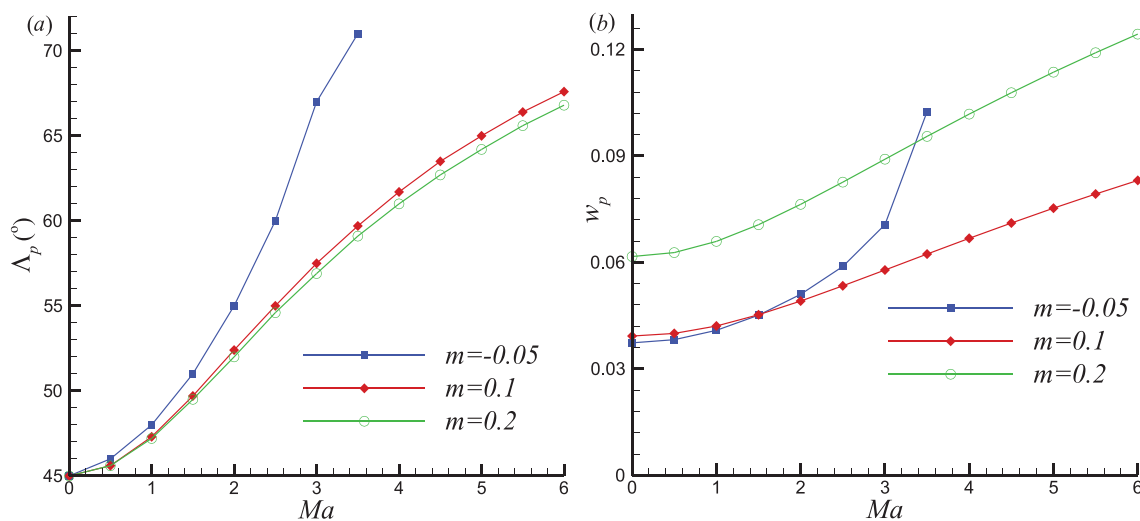


FIG. 16. Variations of cross-flow peak against Mach number ($T_w = T_{ad}$). (a) Local swept angle and (b) peak of cross-flow maximum.

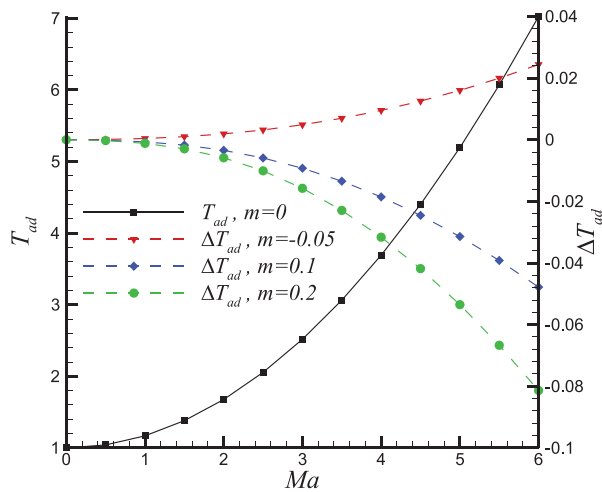


FIG. 17. Variation of adiabatic wall temperature against Mach number ($\Lambda = 45^\circ$). ΔT_{ad} is defined as $T_{ad,m \neq 0} - T_{ad,m=0}$, which represents the wall-temperature difference between 3D and 2D boundary layers.

In a compressible boundary layer under a favorable pressure gradient on a heated wall, an overshoot of streamwise velocity will appear due to the effect of thermal buoyancy.^{29,30} The existence of velocity overshoot results directly in the emergence of a new unstable mode.⁴⁰ The velocity profiles with an overshoot region in current model flow are presented in Fig. 21. In the 2D case, the maximum of streamwise velocity is about 5% larger than that in the external flow. With the increase in local swept angle, crossflow gets stronger and the overshoot gets flatter. It is believed that the momentum transfer from streamwise flow to crossflow accounts for the decrease in the overshoot as well as the whole profile of streamwise velocity.

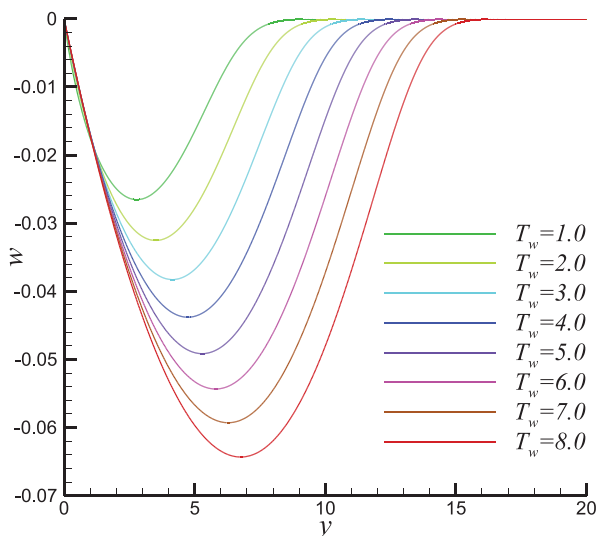


FIG. 18. Cross-flow velocity profiles at various wall temperatures ($m = 0.1$, $\Lambda = 45^\circ$, $Ma = 6.0$).

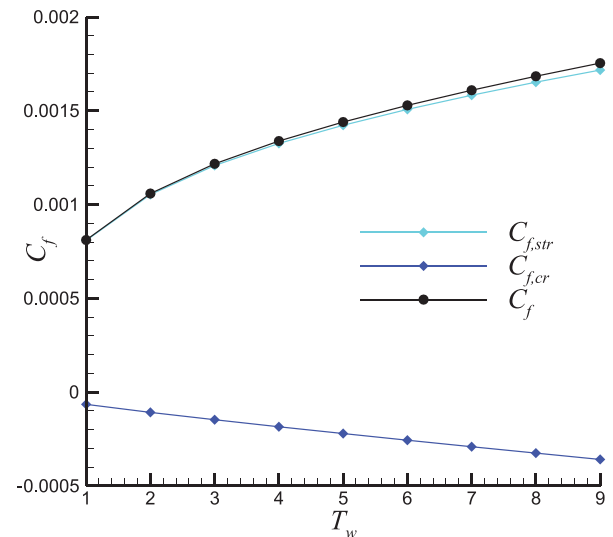


FIG. 19. Variation of skin-friction coefficient against wall temperature ($Ma = 6.0$, $m = 0.1$, $\Lambda = 45^\circ$).

IV. CONCLUSIONS

A flow model of 3D compressible boundary layer has been built on a flat plate in terms of pressure gradient and swept angle. Under the assumption of local similarity, the governing equations are reduced to an ODE system, which is capable of yielding typically C-shape crossflow. The model is an extension of classical FSC flow from incompressible condition to compressible condition. Besides, the model is also an extension of classical compressible boundary layer with self-similar solution from two dimension to three dimension.

In the presented model, a negative distribution of crossflow is obtained under a favorable pressure gradient while positive

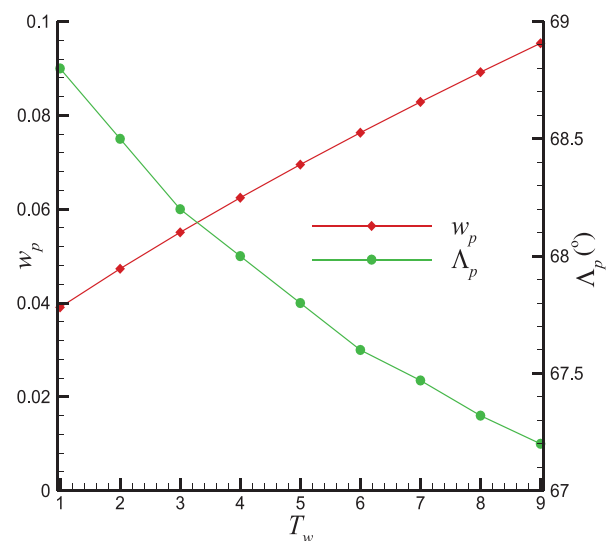


FIG. 20. Cross-flow peak at various wall temperatures ($Ma = 6.0$, $m = 0.1$).

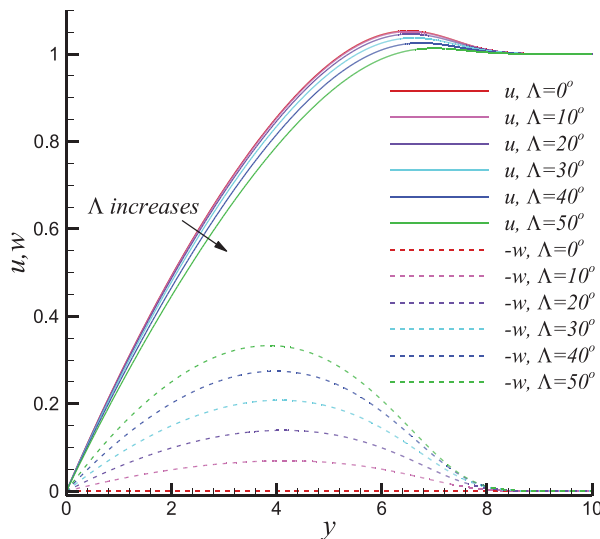


FIG. 21. Velocity profiles on a heated wall at various local swept angles ($Ma = 6.0$, $m = 2.0$, $T_w = 15.0$).

distribution under an adverse one. In either case, the crosswise velocity is always pointing to the curvature center of the external-flow streamline. The occurrence of crossflow distorts the distribution of vorticity by generating a streamwise component. As a consequence, helicity appears inside boundary layer with a profile similar to that of the crossflow. Meanwhile, the crossflow also produces a crosswise component of skin friction. Despite small amplitude compared with the streamwise component, the crosswise component results in a moderate deviation of skin friction from the direction of the external flow. The deviation gets larger and larger downstream although the increase is attenuated due to the decrease in the local swept angle. The crossflow strength increases downstream more and more slowly for the same reason.

The parameters in the model provide quite lot varieties in the boundary-layer characteristics. The pressure-gradient parameter m and the local swept angle Λ are of most importance in controlling crossflow. Larger absolute values of m , indicating stronger pressure gradients, tend to create stronger crossflows and larger deviations of skin friction, while the crossflow strength varies with the local swept angle in a non-monotonic manner. There exists a swept angle Λ_p at which the crossflow strength reaches a peak. As a result, the crosswise skin-friction coefficient varies non-monotonically as well with swept angle and peaks at an angle differing slightly from Λ_p . In addition, Λ_p is found to be exactly 45° in an incompressible boundary layer, and it gets larger at higher Mach numbers. The crossflow strength is revealed to increase as well with increasing Mach number. It is found that heating the wall can also make the crossflow stronger. On a highly heated wall, the existence of crossflow tends to flatten the overshoot in the profile of streamwise velocity. Three general facts are found to exist. First, the variation of crosswise skin friction is revealed to be in accordance with that of the crossflow strength. Second, the crossflow-related properties are more sensitive to m under an adverse pressure gradient. Third, the effect of crossflow on the thermal properties of boundary layer is limited.

The presented boundary layer may serve as a model to investigate fundamentals about crossflow effects in compressible conditions. In particular, the current model could aid to gain more understanding about cross-flow instability or to develop crossflow-induced transition models based on local variables.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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